Siddhant Agarwal

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Education

The University of Austin at Texas — Austin, USA
Ph.D. in Computer Science — *August 2022 - May, 2027 (expected)*
Advisors: Professor Amy Zhang and Professor Peter Stone

Indian Institute of Technology Kharagpur — Kharagpur, India
B.Tech. (Hons.) + M.Tech. (Dual Degree) in Computer Science and Engineering — *July 2017 - June 2022*
Department Rank 1 (GPA 9.90/10.00)

Fields of Interest

Unsupervised RL, GCRL, Representation Learning, Robotics, LLM Finetuning & Reasoning

Publications

(* denotes equal contribution, † denotes equal advising)

Meta-Reinforcement Learning with Zero-Shot Reinforcement Learning
Jake Grigsby, **Siddhant Agarwal***, Yu Lei*, Leonidas Varveropoulos*, Yuke Zhu
40th Annual Conference on Neural Information Processing Systems (NeurIPS) 2026

Unsupervised RL: A Survey and Unified Perspective
Siddhant Agarwal*, Caleb Chuck*, Harshit Sikchi*, Jiaheng Hu, Max Rudolph, Scott Niekum†, Peter Stone†, Amy Zhang†
Submitted *Journal for Machine Learning Research*
RLC Workshop on Reinforcement Learning Beyond Rewards, 2025

Regularized Latent Dynamics Prediction is a Strong Baseline For Behavioral Foundation Models
Pranaya Jajoo, Harshit Sikchi*, **Siddhant Agarwal***, Amy Zhang, Scott Niekum, Martha White
14th International Conference on Learning Representations, 2026
RLC Workshop on Reinforcement Learning Beyond Rewards, 2025

RL Zero: Direct Policy Inference from Language Without In-Domain Supervision
Harshit Sikchi*, **Siddhant Agarwal***, Pranaya Jajoo*, Samyak Parajuli*, Caleb Chuck*, Max Rudolph*, Peter Stone†, Amy Zhang†, Scott Niekum†
39th International Conference on Neural Information Processing Systems(NeurIPS), 2025

Proto Successor Measure: Representing the space of all possible solutions of Reinforcement Learning
Siddhant Agarwal*, Harshit Sikchi*, Peter Stone†, Amy Zhang†
42th International Conference on Machine Learning (ICML), 2025

Reinforcement Learning Within the Classical Robotics Stack: A Case Study in Robot Soccer
Adam Labiosa*, Zhihan Wang*, **Siddhant Agarwal**, William Cong, Geethika Hemkumar, Abhinav Narayan Harish, Benjamin Hong, Josh Kelle, Chen Li, Yuhao Li, Zisen Shao, Peter Stone, Josiah P. Hanna
IEEE Conference on Robotics and Automation (ICRA), 2025

Robot Air Hockey: A Manipulation Testbed for Robot Learning with Reinforcement Learning
Caleb Chuck*, Carl Qi*, Michael J. Munje*, Shuozhe Li*, Max Rudolph*, Chang Shi*, **Siddhant Agarwal***, Harshit Sikchi*, Abhinav Peri, Sarthak Dayal, Evan Kuo, Kavan Mehta, Anthony Wang, Peter Stone, Amy Zhang, Scott Niekum,
ICRA Workshop Manipulation Skills, 2024

f-Policy Gradients: A General Framework for Goal Conditioned RL using f-Divergences
Siddhant Agarwal, Ishan Durugkar, Peter Stone, Amy Zhang
37th International Conference on Neural Information Processing Systems(NeurIPS), 2023

Behavior Predictive Representations for Generalization in Reinforcement Learning
Siddhant Agarwal, Aaron Courville, Rishabh Agarwal
The Multi-Disciplinary Conference on Reinforcement Learning and Decision Making (RLDM), 2022, NeurIPS 2021 EcoRL and DeepRL workshops

Reinforcement Explanation Learning
Siddhant Agarwal, Owais Iqbal, Sree Aditya Buridi, Madda Manjusha, Abir Das
NeurIPS 2021 workshop on eXplainable AI approaches for debugging and diagnosis [Oral]

Poisoned Classifiers are not only backdoored, they are fundamentally broken
Mingie Sun, **Siddhant Agarwal**, Zico Kolter
ICLR 2021 workshop on Security and Safety in Machine Learning Systems

Learning to Deceive Knowledge Graph Augmented Models via Targeted Perturbations
Mrigank Raman, Aaron Chan*, **Siddhant Agarwal***, Peifeng Wang, Hansen Wang, Sungchul Kim, Ryan Rossi, Handong Zhao, Nedim Lipka, Xiang Ren
International Conference on Learning Representations (ICLR), 2021

Traffic Sign Classification using Hybrid HOG-SURF Features using Convolutional Neural Networks
Rishabh Madan*, Deepank Agrawal*, Shreyas Kowshik*, Harsh Maheshwari*, **Siddhant Agarwal***, Debashish Chakraborty (* denotes equal contribution-dice roll)
International Conference on Pattern Recognition Applications and Methods, Prague 2019

Experience

ML Research Intern May 2026-August 2026
Nuro *Mountain View*

Utilized energy-based modeling to allow flow model to leverage suboptimal data for reducing long tail generations during trajectory prediction in autonomous driving.

Applied Scientist Intern May 2024-April 2025
Amazon *New York*

Developed a principled exploration strategy to collect online samples to provably improve offline RL datasets by reducing the sub-optimality gap in the offline dataset. Integrated this algorithm to improve the Supply Chain buying policy.

Data Engineering Intern June 2021 - July 2021
Goldman Sachs *India*

Developed a python client to use the services of an ETL platform to quickly transfer data from pandas dataframe to a MemSQL store. Created RESTful server endpoints for the ETL platform and accessed them from the client.

Research Intern May 2020 - October 2020
LocusLab, Carnegie Mellon University, ***Advisor: Zico Kolter*** *Pittsburg, USA*

Showed that *backdoored classifiers* are fundamentally broken and can be attacked by anyone rather than only the adversary.

Research Intern March 2020 - May 2020
INKLab, University of Southern California, ***Advisor: Xiang Ren*** *Los Angeles, USA*

Trained DQN policies to heavily perturb KGs without changing the downstream performance of the different KG models. Was able to deceive MHGRN, GRN and RN (on CSQA and OBQA) and KGCN and RippleNet (on MovieLens and Music).

Autonomous Ground Vehicles Research Group March 2018 - March 2022
Indian Institute of Technology Kharagpur *Kharagpur, India*

Worked to develop a category 4 autonomous ground vehicle. Developed traffic sign classification system for Indian traffic signs, lane segmentation systems, planning using Frenet Serret Local Planner and Time Elastic Bands and Localization systems using Extended Kalman Filters.

Invited Talks

The Causal Reinforcement Learning Workshop, RL Conference 2025
Learning Generalist Agents using Causal MDP Abstractions

The International Summer School on Situational Awareness in Cognitive Technologies 2021
with Prof. Abir Das, hands-on session with on explainable AI

Competitions

RoboCup Standard Platform League *First Place in Challenge Shield, 2024* 2023, 2024

Prepared a team of 5 Nao robots to compete in a soccer tournament which involved developing the basic motion system like walking and kicking along with localization to estimate the agent's position in the field. Used RL to train behavior strategies in the game to score goals.

Bosch Traffic Sign Detection, Inter IIT Tech Meet *Bronze* 2021

Developed a web-based tool to add real-life data augmentations on a traffic sign dataset, train classifiers, and debug them. The tool helps the users to visualize the mistakes of the model and debug the model.

Intelligent Ground Vehicles Competition, Oakland University, Michigan, *Runners Up* 2018 & 2019

Developed an autonomous ground vehicle for a constraint environment using simple Computer Vision and Image Processing techniques for Lane navigation and Obstacle avoidance, TEB local planner for planning. First team to qualify.

Teaching Experiences and Organizing Services

Graduate Teaching Assistant

- CS394R Reinforcement Learning, The University of Texas at Austin 2024
- CS343 Artificial Intelligence, The University of Texas at Austin 2022
- CS60010 Deep Learning, IIT Kharagpur 2022
- CS60077 Reinforcement Learning, IIT Kharagpur 2021

"Oops, I Erred", Workshop on Collecting, Curating and Using Imperfect Data for Realistic Scenarios at CoRL, Austin, *Organizer* 2026

Member of the organization team for the workshop, worked on reviewer allotment, accepted paper list and overall management of the workshop.

Workshop on Reinforcement Learning Beyond Rewards: Towards Scalable General-Purpose Agents at The RL Conference, Montreal, *Organizer* 2026

Member of the core organization team for the workshop, worked on inviting speakers, schedule, website, reviewer allotment, accepted paper list, spotlights and overall management of the workshop.

Workshop on Reinforcement Learning Beyond Rewards: Ingredients for Developing Generalist Agents at The RL Conference, Edmonton, *Organizer* 2026

Member of the core organization team for the workshop, worked on inviting speakers, schedule, website, accepted paper list, spotlights and overall management of the workshop.

Workshop on Reinforcement Learning Beyond Rewards at The RL Conference, Amherst, *Organizer* 2024

Member of the core organization team for the workshop, helped to ideate the workshop, worked on the invited speaker invitations, schedule, website, accepted paper list, spotlights and overall management of the workshop.

Dynamic Neural Networks meet Computer Vision Workshop at CVPR, *Volunteer & Reviewer* 2021

Volunteered to help the organizers with the logistics of the workshop, and created the workshop posters.

Reviewer

ICLR 2026, NeurIPS 2025 (**Top Reviewer**), ICML 2025, ICLR 2025, RLC 2025, NeurIPS 2024, ICLR 2024, NeurIPS 2023 Goal Conditioned Reinforcement Learning workshop, NeurIPS 2023, CVPR 2021 workshop on Dynamic Neural Networks meet Computer Vision

Achievements

Top Reviewer, NeurIPS 2025

Most Creative Poster at the Texas Robotics Symposium 2023

Graduate Scholarship from ECE Dept, UT Austin 2023

Institute Silver Medal for securing Rank 1 in the Department of Computer Science and Engineering 2022

GATE Scholarship for Teaching Assistantship 2021

IUSSTF - Viterbi Scholarship - ***Only ten students selected from India*** 2020

Class of 1970 Alumni (US) Association Prize for ***best student in order of merit*** **among third year B.Tech.(Hons.)/B.Arch.(Hons.) courses securing highest CGPA** 2019

IIT Kharagpur Alumni (California Chapter) Award for being the ***best student in order of merit*** **among third year B.Tech.(Hons.)/B.Arch.(Hons.) courses securing highest CGPA** 2019

Student Par Excellence Award by Computer Science & Engineering Dept 2018

KVPY Fellowship Award - ***Among the top 0.6% of the applicants*** 2017

OutReach

Texas Robot Parade 2023, 2024, 2025

Demonstrated the Naos from our RoboCup Standard Platform League Team that plays soccer in a competitive tournament.

Directed Reading Program, ***Mentor*** 2023

Organize a reading group on reinforcement learning for undergraduate students to get them involved in RL research